A Multi-Site Label Trap in Shape-Based Endometriosis Detection

What female pelvic-organ shape does (and does not) encode

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424
women across two public MRI datasets

78% vs 15%
endometrioma annotation prevalence, site D1 vs D2

0.826
AUC within site D2 (95% CI 0.67–0.94)

0.274
transfer AUC D1→D2 (CI entirely below 0.5)

Three questions

Does uterine shape encode age, and is it the uterus or the fibroids talking?

Can shape predict which patients carry an endometrioma annotation, and does that survive a site change?

How stable are shape descriptors when a different rater segments the organ?

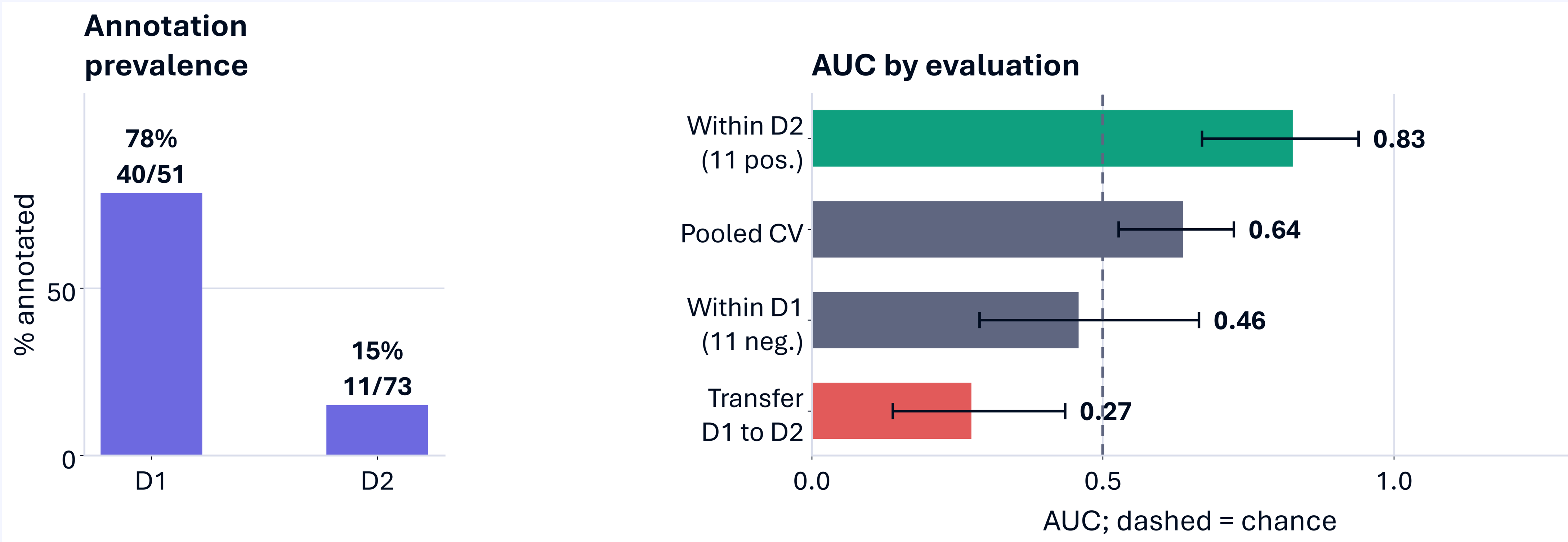
Data & methods

UMD: 300 women, ages 21–86.
UT-EndoMRI: 124 patients, two sites, three raters at D1.

XGBoost (gradient-boosted trees), subject-level 5-fold CV, bootstrap 95% CIs.

Size descriptors grow with the organ (volume, area, overall dimensions); scale-free shape captures its form regardless of size (sphericity, elongation, aspect ratios).

The label trap: one model, four verdicts



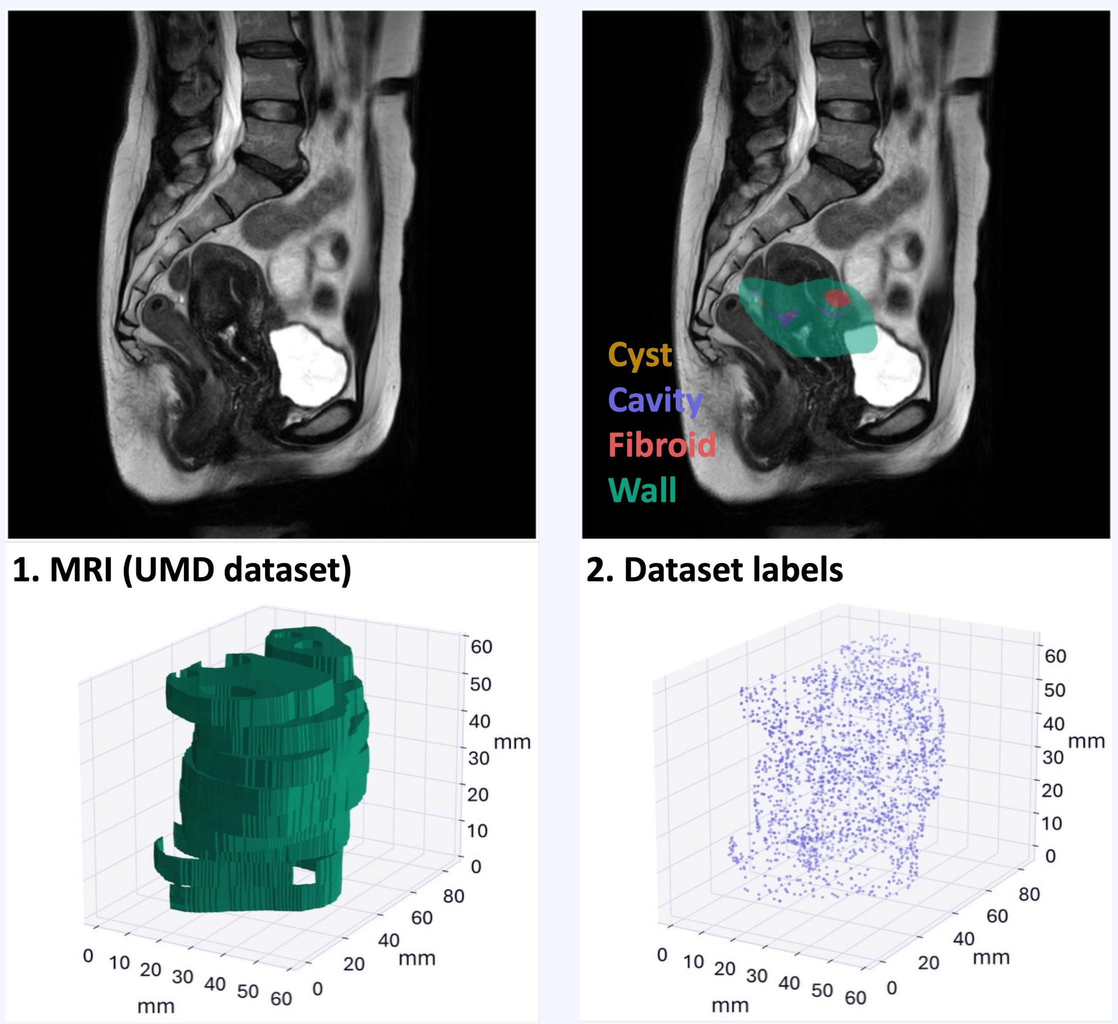
The endometrioma label marks the presence of an annotation file. D1 annotated endometriomas in 78% of its patients, while D2 annotated only 15%. These are annotation habits, not related to true prevalence.

Within D2 the annotation is predictable from shape: AUC 0.826 (95% CI 0.67–0.94, permutation p = 0.004).

Pooled cross-validation conflates site with pathology (0.638); transfer D1→D2 collapses to 0.274 with the CI entirely below 0.5.

Any multi-site study where annotation completeness tracks the target risks this confound. Report per-site prevalence and cross-site transfer.

The pipeline



1. MRI (UMD dataset)
2. Dataset labels
3. Marching-cubes mesh
4. 2,048 surface points

Before you pool

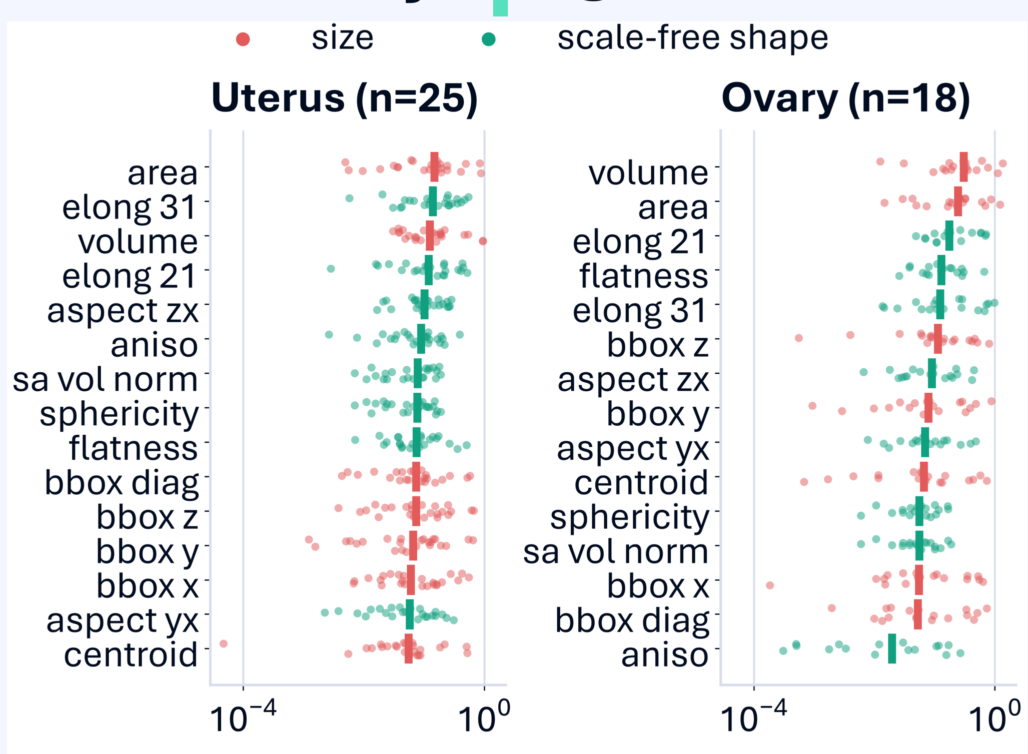
Check label prevalence per site.

Test cross-site transfer.

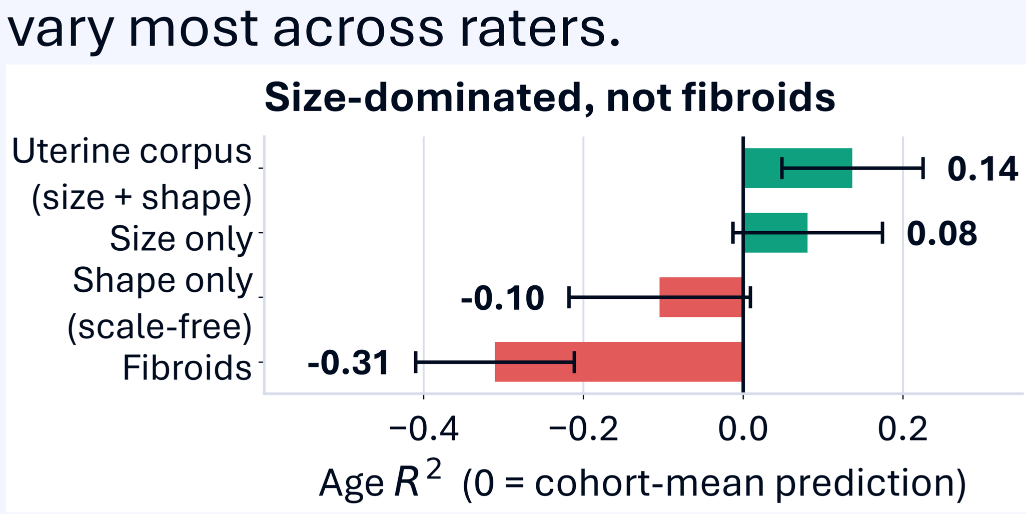
Report a size control next to any shape biomarker.

Rater stability

Age: size-driven

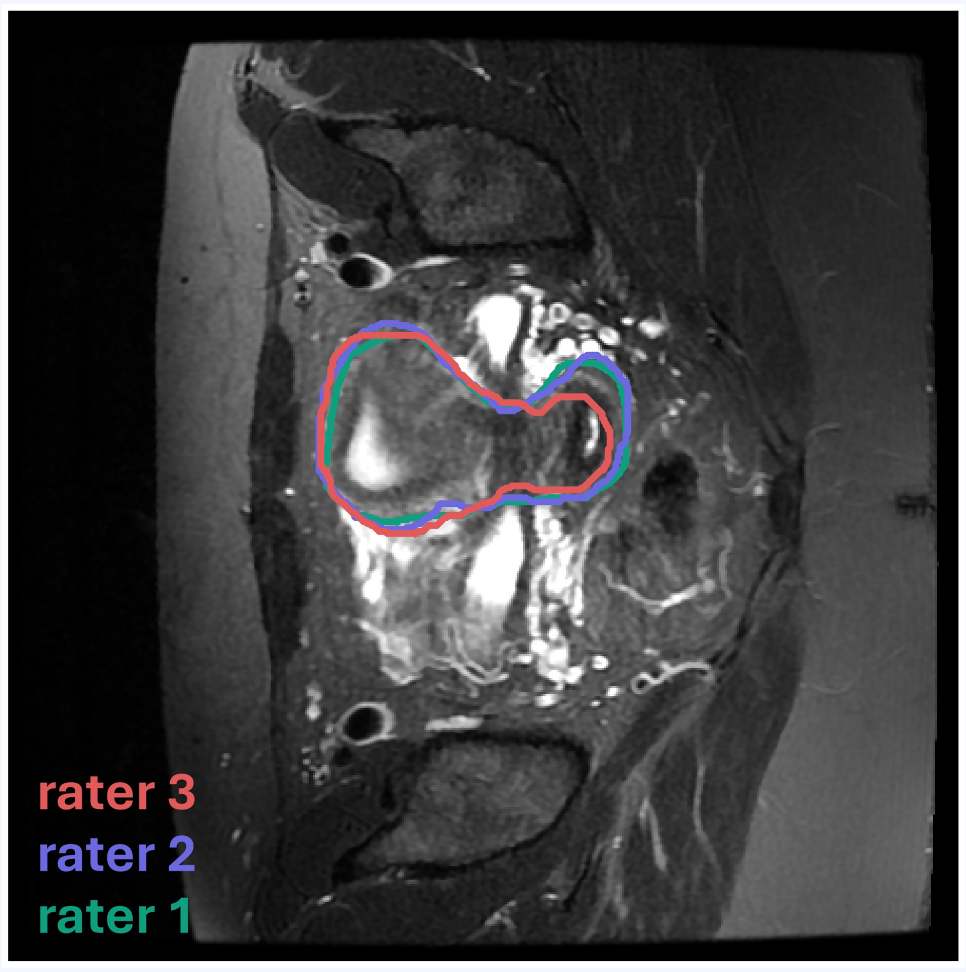


Averaged over all descriptors, shape is more rater-stable than size; the two groups overlap, and volume and area vary most across raters.



MAE 8.9 yr vs 10.2 baseline; scale-free shape alone falls below the mean; fibroids carry nothing.

Three raters, one uterus



Three raters outline the same uterus (site D1); the boundary is where they disagree.

Limitations

Modest cohorts; annotation labels; 11 positives behind the headline AUC; raters from one site (25 uterus / 18 ovary subjects).

Code, data pipeline, and interactive figures
github.com/TIO-IKIM/shape-demographics-capi
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Data: UMD (Pan et al., Sci Data 11:410, 2024, CC BY 4.0)
UT-EndoMRI (Liang et al., Sci Data 12:1292, 2025).

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